

CO2_AT2

Weather Monitoring System – Closest Pair Using Brute Force

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Aim:

To find the nearest pair of weather stations using the brute-force closest pair algorithm and analyze its time complexity.

Procedure

1. Define the coordinates of all weather stations.
2. Compare every station with every other station.
3. Calculate the Euclidean distance between each pair.
4. Store the minimum distance and the corresponding pair.
5. Display the closest pair and minimum distance.
6. Analyze the time complexity.

Algorithm

Step 1: Start.

Step 2: Read the coordinates of weather stations.

Step 3: Initialize the minimum distance as infinity.

Step 4: Compare every pair of stations using nested loops.

Step 5: Calculate the Euclidean distance for each pair.

Step 6: Update the minimum distance if a smaller distance is found.

Step 7: Display the closest pair and minimum distance.

Step 8: Stop.

Code & Output

main.py	Output
<pre>1 import math 2 3 stations = [(2, 8), (5, 6), (9, 4), (4, 7), (8, 3)] 4 5 min_dist = float('inf') 6 closest_pair = () 7 8 for i in range(len(stations)): 9 for j in range(i + 1, len(stations)): 10 x1, y1 = stations[i] 11 x2, y2 = stations[j] 12 13 dist = math.sqrt((x2 - x1)**2 + (y2 - y1)**2) 14 15 if dist < min_dist: 16 min_dist = dist 17 closest_pair = (stations[i], stations[j]) 18 19 print("Closest Stations:", closest_pair) 20 print("Minimum Distance:", round(min_dist, 2)) 21 print("Time Complexity = $\Theta(n^2)$")</pre>	<pre>Closest Stations: ((5, 6), (4, 7)) Minimum Distance: 1.41 Time Complexity = $\Theta(n^2)$ === Code Execution Successful ===</pre>

Analysis

- Number of weather stations = 5
- Total pairs compared = $5 \times 4 / 2 = 10$
- Every pair is checked exactly once.
- Time Complexity = $\Theta(n^2)$

Result

The brute-force closest pair algorithm successfully identifies the nearest weather stations by evaluating all possible pairs. Although it guarantees the correct result, its $\Theta(n^2)$ time complexity makes it inefficient for large-scale weather monitoring systems containing a large number of stations.